

# **AJOL Journal Analytics Dashboard: Publication Trends, Open Access Distribution, and Journal Performance Analysis (2020–March 2024)**

## **1. Project Overview and Objective**

### **Project Overview**

This project focuses on analyzing publication data from African Journals Online (AJOL) between 2020 and March 2024. The dataset was cleaned and transformed using Excel and Power Query, followed by data modelling and interactive dashboard development in Power BI. The dashboard provides insights into publication trends, journal performance, open access distribution, geographical coverage, and journal quality based on the Journal Publishing Practices and Standards (JPPS) classification.

### **Objective**

The primary objective of this project is to analyse publication trends, journal performance, Open Access distribution, geographical coverage, and JPPS journal quality using Excel and Power BI. The project also demonstrates data preprocessing, modelling, and interactive dashboard development to support data-driven decision-making in scholarly publishing.

## **2. Data Sources**

### **Dataset Title:**

AJOL Dataset: Structured Metadata of Articles and Journals Indexed in African Journals Online

### **Original Source:**

[Zenodo Dataset – AJOL Dataset: Structured Metadata of Articles and Journals Indexed in African Journals Online](#)

### **DOI:**

10.5281/zenodo.15383374

### **Dataset Version:**

Version 2 (Published: May 2025)

### **Dataset Timeline:**

Articles published between **2020 and March 2024** (filtered for this project).

### **Dataset Size:**

Approximately

- 50,000 articles
- 739 journals
- 153,000 authors
- 221,000 keywords

### **Domain:**

Scholarly Publishing / Research Analytics / Bibliometrics

### **Dataset Description:**

The dataset contains structured metadata of journals and research articles indexed in African Journals Online (AJOL). It includes information related to journals, publications, authors, keywords, countries, JPPS ratings, and Open Access status. The dataset was obtained from Zenodo and filtered to include publications from 2020–March 2024 for analysis.

### **Dataset Files Used:**

- **ajol\_journals.csv** – Journal information (country, JPPS status, Open Access, ISSN)
- **ajol\_pub.csv** – Article metadata
- **ajol\_pub\_author.csv** – Author information
- **ajol\_pub\_keyword.csv** – Keywords for each article

## **3. Objectives**

- To analyze annual publication trends between 2020 and March 2024.
- To identify the journals publishing the highest number of research articles.
- To examine the distribution of Diamond and Non-Diamond Open Access journals.
- To analyze country-wise publication output across African nations.
- To evaluate journal quality using the JPPS classification.

- To build an interactive dashboard for exploring publication and journal performance.

## 4. Attribute (Column / Feature) Details

| Attribute Name     | Data Type | Description                                                                                                  |
|--------------------|-----------|--------------------------------------------------------------------------------------------------------------|
| Article_ID         | Integer   | Unique identifier assigned to each published article.                                                        |
| Journal_ID         | Integer   | Unique identifier assigned to each journal.                                                                  |
| Journal_Title      | Text      | Title or name of the journal in which the article is published.                                              |
| Publication_Year   | Integer   | Year in which the article was published.                                                                     |
| Publication_Date   | Date      | Complete publication date of the article.                                                                    |
| Author             | Text      | Name of the article author(s).                                                                               |
| Keyword            | Text      | Keywords associated with the article for indexing and search.                                                |
| Country            | Text      | Country where the journal is published.                                                                      |
| Open Access Status | Text      | Indicates whether the journal follows the <b>Diamond</b> or <b>Non-Diamond</b> Open Access publishing model. |
| JPPS_Status        | Text      | Journal Publishing Practices and Standards (JPPS) quality classification of the journal.                     |
| DOI                | Text      | Digital Object Identifier (DOI) assigned to uniquely identify the research article.                          |

## 5. Tools & Technologies

### Microsoft Excel

- Data Cleaning
- Data Transformation
- Filtering & Sorting
- Duplicate Removal

### Power Query

- Data Import
- Data Cleaning
- Data Type Conversion
- Merge Queries
- Remove Errors

- Remove Blank Records

## **Power BI**

- **Data Modelling**
- **Relationships**
- **DAX Measures**
- **Interactive Dashboard**
- **Report Creation**

## **6. Data Pre-Processing (Excel / Power Query)**

### **Tasks Performed**

Data cleaning and transformation were performed using Power Query Editor, including duplicate removal, null handling, data type conversion, filtering, and column standardization.

### **Data Cleaning**

- Removed duplicate records.
- Removed null values from important columns.
- Corrected inconsistent data formats.
- Fixed text formatting issues.
- Converted appropriate data types.

### **Data Transformation**

- Filtered publication years (2020–March 2024).
- Renamed columns for better readability.
- Standardized country and journal names.
- Converted Open Access values into descriptive categories.

### **Data Preparation**

- Split the dataset into Fact and Dimension tables.
- Established unique identifiers.

- Loaded cleaned data into Power BI.

## **7. Data Modelling and DAX (Power BI)**

### **Data Model**

A Star Schema data model was implemented by connecting the Publication table (Fact Table) with Journal, Author, and Keyword dimension tables using one-to-many relationships.

### **Fact Table**

- Publication Table

### **Dimension Tables**

- Journal
- Author
- Keyword

Relationships were created using **one-to-many cardinality** between the fact table and dimension tables.

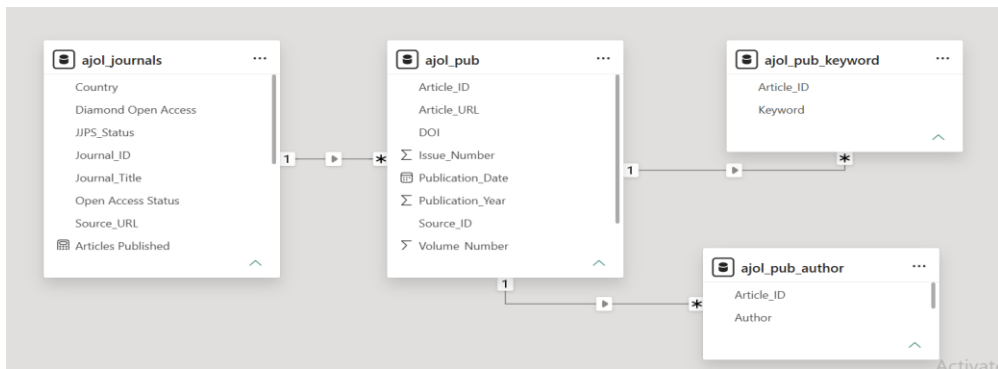
### **Calculated Columns Created**

- Open Access Status
- Publication Year
- JPPS Status (formatted)

### **Measures Created**

- Total Articles
- Total Journals
- Total Authors
- Total Keywords
- Countries Covered
- Diamond Journals

- Article Count



## 8. Analysis and Visualizations (Power BI)

### Dashboard Features

The dashboard provides interactive filtering using slicers, dynamic cross-filtering, tooltips, and KPI cards for enhanced data exploration.

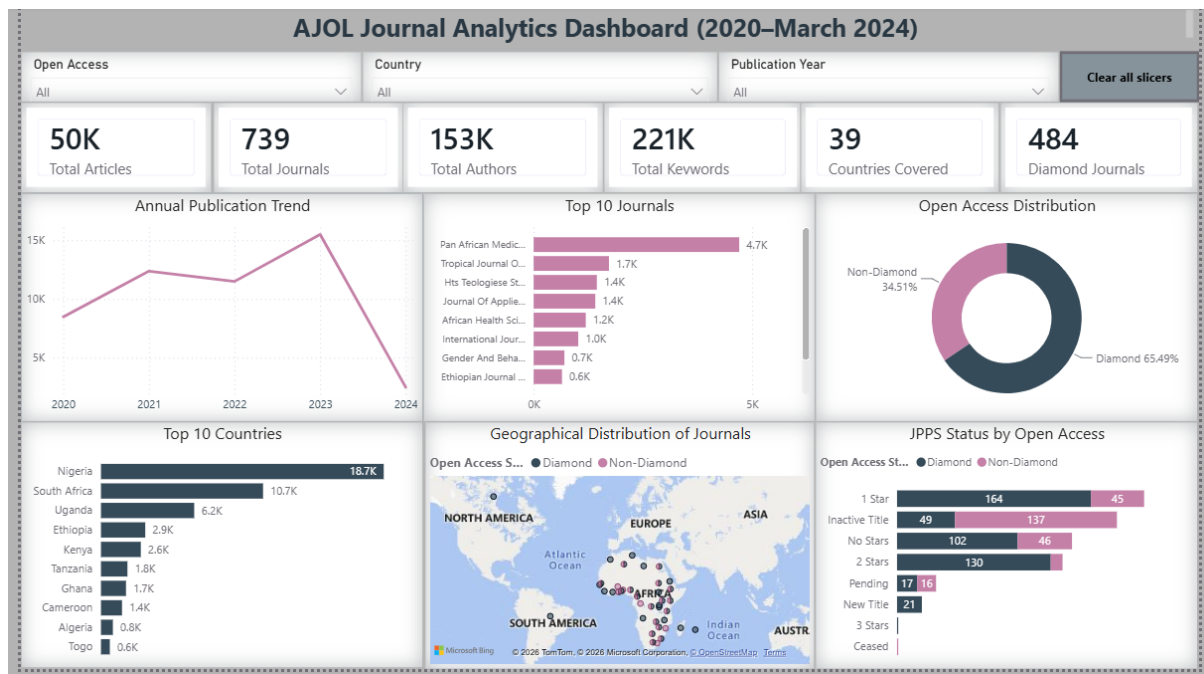
The dashboard contains the following visualizations:

1. KPI Cards
  - Total Articles
  - Total Journals
  - Total Authors
  - Total Keywords
  - Countries Covered
  - Diamond Journals
2. Annual Publication Trend (Line Chart)
3. Top 10 Journals by Article Count (Bar Chart)
4. Open Access Distribution (Donut Chart)
5. Top 10 Countries by Article Count (Bar Chart)
6. Geographical Distribution of Journals (Map)
7. Open Access by JPPS Status (Stacked Bar Chart)

### Interactive Features

- Publication Year Slicer

- Country Slicer
- Open Access Slicer
- Clear All Slicers Button
- Dynamic Cross Filtering



## 9. Insights & Conclusions

### Visual 1 – KPI Cards

#### Insight 1

##### Descriptive Analysis

- The dataset contains **50,000 articles**, **739 journals**, **153,000 authors**, **221,000 keywords**, and publications from **39 countries**.

##### Diagnostic Analysis

- The large volume of publications indicates extensive scholarly coverage across multiple disciplines and countries.

##### Predictive Analysis

- Continued journal participation is expected to increase publication volume in future years.

##### Prescriptive Analysis

- Expand journal participation and maintain regular metadata updates to improve research visibility.

## Visual 2 – Annual Publication Trend

### Insight 2

#### Descriptive Analysis

- Publications increased steadily from **2020 to 2023**, with **2023 recording the highest publication count**.

#### Diagnostic Analysis

- The decline observed in **2024** is due to the dataset containing records only until **March 2024**.

#### Predictive Analysis

- Publication output is expected to remain consistent if current trends continue.

#### Prescriptive Analysis

- Update the dashboard periodically as new publication records become available.

## Visual 3 – Top 10 Journals

### Insight 3

#### Descriptive Analysis

- **Pan African Medical Journal** published the highest number of research articles.

#### Diagnostic Analysis

- A small number of journals contribute a significant proportion of AJOL publications.

#### Predictive Analysis

- These journals are likely to continue leading publication output.

#### Prescriptive Analysis

- Encourage knowledge sharing between high-performing and emerging journals.

## Visual 4 – Open Access Distribution

### Insight 4

#### Descriptive Analysis



- **Diamond Open Access journals account for 65.49%, while 34.51% are non-Diamond.**

#### **Diagnostic Analysis**

- Diamond Open Access is the dominant publishing model across AJOL journals.

#### **Predictive Analysis**

- Diamond Open Access adoption is expected to continue increasing.

#### **Prescriptive Analysis**

- Continue supporting sustainable Diamond Open Access publishing initiatives.

### **Visual 5 – Top 10 Countries**

#### **Insight 5**

##### **Descriptive Analysis**

- **Nigeria, South Africa, and Uganda** contribute the highest publication output.

##### **Diagnostic Analysis**

- Research productivity is concentrated within a few African countries.

##### **Predictive Analysis**

- These countries are expected to remain leading contributors.

##### **Prescriptive Analysis**

- Encourage research collaboration and publication support for lower-contributing countries.

### **Visual 6 – JPPS Status by Open Access**

#### **Insight 6**

##### **Descriptive Analysis**

- Most journals are classified as **1 Star** under the JPPS system.

##### **Diagnostic Analysis**

- Only a limited number of journals have achieved higher JPPS ratings.

##### **Predictive Analysis**

- Without quality improvement initiatives, higher-rated journal growth may remain limited.

## Prescriptive Analysis

- Strengthen editorial practices, peer review, and publishing standards to improve JPPS ratings.

## 10. Conclusion

The integration of **Excel**, **Power Query**, and **Power BI** enabled efficient data cleaning, transformation, modelling, and visualization of AJOL publication data. The dashboard successfully highlights publication trends, journal performance, open access distribution, geographical coverage, and journal quality. The developed dashboard provides an effective decision-support tool for publishers, researchers, librarians, and policy makers by enabling data-driven evaluation of publication performance, Open Access adoption, and journal quality across African scholarly publishing.

## References

Alonso-Álvarez, P. (2025). *AJOL Dataset: Structured Metadata of Articles and Journals Indexed in African Journals Online* (Version 2). Zenodo.  
<https://doi.org/10.5281/zenodo.15383374>